MODULE:-2 ASSI=1

1. CREATE A TABLE LISTING 3 TYPES OF PRINTERS, 2 TYPES OF SCANNERS, AND 2 TYPES OF WEBCAMS, AND FOR EACH, WRITE ONE REAL-LIFE USE CASE (FOR EXAMPLE: 'INKJET PRINTER – PRINTING COLLEGE ASSIGNMENTS').

Types of Printers, Scanners, and Webcams

Category	Type	Description	Real-Life Use Case
Printer	Inkjet Printer	An inkjet printer produces images and text by spraying tiny droplets of liquid ink onto paper. It is widely used because it can produce high-quality colour prints and is particularly suitable for photographs, graphics, and documents containing colour. Inkjet printers are generally affordable for home and small-office users, although ink replacement can become a significant running cost.	**Printing college assignments** – Students can use an inkjet printer to print assignments, project reports, diagrams, charts, and colour photographs at home or in a college laboratory.
Printer	Laser Printer	A laser printer uses a laser beam, a photosensitive drum, and powdered toner to create printed documents. It is known for fast printing speeds, sharp text, and efficient handling of large volumes of documents. Laser printers are especially common in offices, schools, banks, and other organizations where many pages need to be printed regularly.	**Printing office reports and invoices** – A business can use a laser printer to quickly produce hundreds of pages of reports, bills, forms, and other text-heavy documents with consistent quality.
Printer	3D Printer	A 3D printer creates physical three-dimensional objects from digital designs by depositing or solidifying material layer by layer. Depending on the technology, it can work with materials such as plastic filament, resin, or specialized industrial materials. 3D printing is used for prototyping, education, engineering, manufacturing, healthcare, and product development.	**Creating an engineering prototype** – An engineering student or company can print a physical model of a machine component or product design before manufacturing the final version.

Scanner	**Sheet-Fed Scanner**	A sheet-fed scanner automatically pulls individual sheets of paper through a scanning mechanism. It is designed primarily for documents rather than books or thick objects and is often equipped with an automatic document feeder (ADF), allowing multiple pages to be scanned efficiently.	**Digitizing business documents** – An office can use a sheet-fed scanner to scan a large stack of invoices, receipts, forms, or employee records and convert them into searchable digital files.
Scanner	**Flatbed Scanner**	A flatbed scanner has a flat glass surface on which a document, photograph, book page, or other object is placed. A scanning head moves beneath the glass to capture the material digitally. Flatbed scanners are valued for their versatility and ability to produce high-quality scans, making them useful for both documents and photographs.	**Digitizing old family photographs** – A person can scan printed photographs and save them as digital image files for long-term storage, editing, sharing, or backup.

Webcam	**External USB Webcam**	An external USB webcam is a separate camera that connects to a computer through a USB port. It often provides better positioning, image quality, field of view, and microphone capabilities than basic built-in cameras. It can be mounted on a monitor, tripod, or other suitable surface and is commonly used for professional communication and content creation.	**Participating in a work-from-home video conference** – An employee can use an external USB webcam to provide a clearer and better-positioned video feed during meetings with colleagues or clients.
Webcam	**Built-in Webcam**	A built-in webcam is integrated directly into a device such as a laptop, tablet, or all-in-one computer. It normally provides a convenient camera for video communication without requiring additional hardware. Modern built-in webcams may support features such as HD video, automatic exposure, noise reduction, and facial framing.	**Attending an online class** – A student can use the laptop's built-in webcam to participate in a virtual classroom and communicate face-to-face with teachers and classmates.

2. DRAW A LABELED DIAGRAM SHOWING HOW A TYPICAL HOME PRINTER CONNECTS TO A LAPTOP OR PC, INCLUDING THE CONNECTION INTERFACE USED (USB, WI-FI, OR BLUETOOTH).

HOME PRINTER CONNECTION

LAPTOP / PC

Documents / Photos

Wi-Fi
(Wireless
connection)

WI-FI ROUTER

Home Network

Wi-Fi

HOME PRINTER

Paper / Ink /
Toner

Printed Page

ALTERNATIVE CONNECTION INTERFACES

Laptop / PC ———————— USB Cable ———————— Printer
(Wired)

Laptop / PC)))))) Bluetooth)))))) Printer
(Wireless)

Theory

A printer is an output device that is used to produce text, images, and documents on paper. A home printer can be connected to a laptop or PC using different connection interfaces such as **USB, Wi-Fi, or Bluetooth**.

In a **USB connection**, the printer is directly connected to the laptop or PC using a USB cable. This is a simple and reliable wired connection.

In a **Wi-Fi connection**, the printer and laptop are connected to the same wireless network through a Wi-Fi router. The user can send a document from the laptop to the printer without using any cable.

In a **Bluetooth connection**, the laptop and printer communicate wirelessly over a short distance. This is useful when both devices support Bluetooth and a direct wireless connection is preferred.

Example: If a student wants to print an assignment from a laptop using a Wi-Fi printer, the laptop sends the assignment through the home Wi-Fi network to the printer, and the printer prints the assignment on paper.

3. LIST THREE DIFFERENT CONNECTION INTERFACES (SUCH AS USB, WI-FI, BLUETOOTH) USED BY PERIPHERALS LIKE PRINTERS, SCANNERS, AND WEBCAMS, AND FOR EACH, NAME ONE ADVANTAGE AND ONE DISADVANTAGE.

<EM><STRONG>HINT:</STRONG> THINK ABOUT SPEED, CONVENIENCE, AND COMPATIBILITY.</EM>

CONNECTION INTERFACES USED BY PERIPHERALS

1. Peripherals such as **printers, scanners, and webcams** can connect to a computer using different interfaces. The three common connection interfaces are **USB, Wi-Fi, and Bluetooth.**

Connection Interface	Used By	Advantage	Disadvantage
USB (**Universal Serial Bus**)	Printers, scanners, webcams	**Fast and reliable:** USB provides a stable connection and is generally suitable for transferring large amounts of data, such as high-quality scanned images or webcam video.	**Requires a cable:** The device must be physically connected to the computer, which can limit movement and create cable clutter.
Wi-Fi	Printers, scanners, webcams	**Convenient and wireless:** Devices can connect without cables, and multiple computers or devices can often access a network printer.	**Depends on network quality:** A weak or unstable Wi-Fi connection can cause slow data transfer or connection problems.
Bluetooth	Printers, webcams, some scanners and other peripherals	**Easy short-range wireless connection:** It reduces the need for cables and is convenient for connecting compatible devices.	**Limited range and speed:** Bluetooth generally has a shorter operating range and lower data-transfer capability than a wired USB connection.

4. PICK ANY ONE DEVICE YOU USE AT HOME OR COLLEGE (PRINTER, SCANNER, OR WEBCAM). WRITE STEP-BY-STEP INSTRUCTIONS FOR HOW YOU WOULD CONNECT IT TO YOUR COMPUTER, INCLUDING WHAT CABLE OR WIRELESS METHOD YOU WOULD USE AND ANY SETUP STEPS REQUIRED.

CONNECTING A PRINTER TO A COMPUTER

I would choose a **printer** as the device. I would connect the printer to my computer using a **USB cable** because it is simple, reliable, and easy to set up.

Step-by-Step Instructions

1. **Place the printer properly:**
   Keep the printer on a stable surface near the computer and connect its power cable to an electrical socket.

2. **Turn on the printer:**
   Press the power button and wait for the printer to start. Make sure there is enough paper and ink or toner.

3. **Connect the USB cable:**
   Connect one end of the **USB cable** to the USB port on the printer and the other end to an available USB port on the computer.

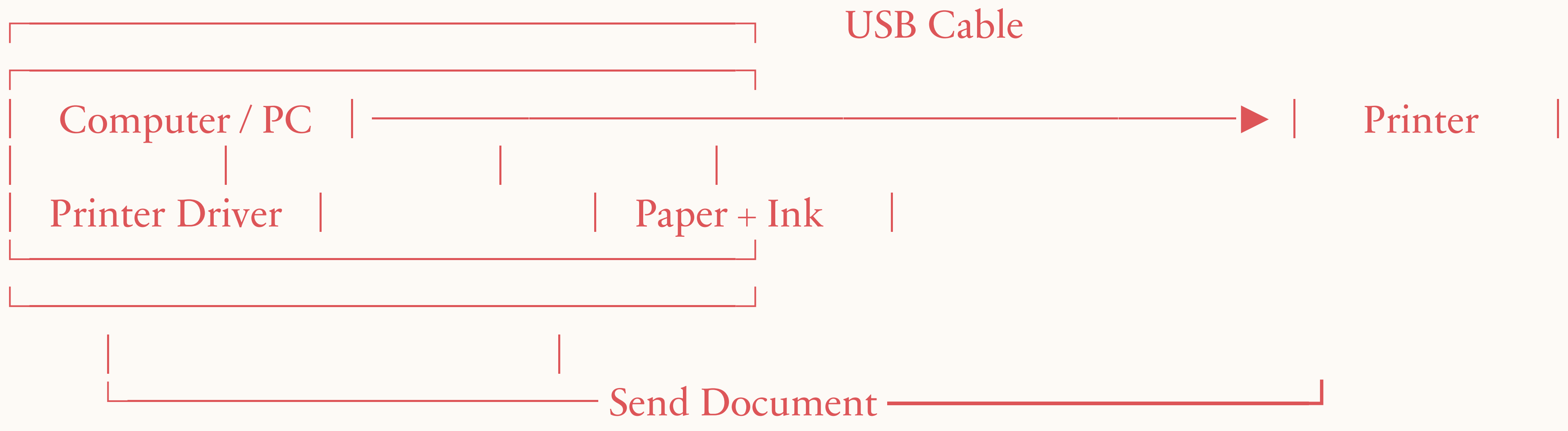
4. **Allow the computer to detect the printer:**
   After connecting the cable, the computer should automatically detect the printer. A notification may appear indicating that the device is being set up.

5. **Install the printer driver:**
   If required, install the correct **printer driver/software** supplied by the printer manufacturer. The driver allows the computer's operating system to communicate properly with the printer.

6. **Add the printer:**
   Open the computer's **Printer Settings** and check whether the connected printer appears in the list of available printers. Select and add it if necessary.

7. **Set it as the default printer:**
   If it is the printer used most frequently, select the option to make it the **default printer**.

8. **Print a test page:**
   Open a document and select **Print**. Choose the connected printer and print a test page to confirm that the connection is working correctly.



- **A USB CONNECTION IS A CONVENIENT CHOICE FOR CONNECTING A PRINTER DIRECTLY TO A COMPUTER. AFTER CONNECTING THE CABLE, INSTALLING THE REQUIRED DRIVER, ADDING THE PRINTER, AND PRINTING A TEST PAGE, THE PRINTER IS READY TO USE.**

THANK YOU